
Nonlinear Plasma Waves from a ZZ -Extended Spin Chain: Modulational Instability and the Nonlinear Schrödinger Limit

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Abstract

Linear plasma-wave propagation can be encoded in the single-excitation subspace of a spin chain whose hopping and staggered field reproduce the relativistic dispersion $\omega_k = \pm\sqrt{\omega_p^2 + c^2 k^2}$ [Sundar et al., Phys. Rev. Applied 25, 024077 (2026)]. We show that augmenting this chain with a nearest-neighbour ZZ interaction drives the model into the nonlinear regime and, in the classical (coherent-state) limit, maps it onto the focusing nonlinear Schrödinger equation (NLSE). Through a bosonization and a continuum envelope reduction we identify the ZZ coupling as the source of the cubic self-phase-modulation term, with nonlinearity coefficient $q = -2\epsilon$. We then demonstrate that the resulting modulational instability — the exponential growth of a small sideband on a coherent carrier — is reproduced quantitatively: a lattice mean-field (Hartree) reduction of the spin model yields growth rates matching the NLSE prediction to within $\sim 15\%$ across a four-point sweep of the ZZ strength, while a full- 2^N -Hilbert-space statevector emulation on 12 qubits confirms the instability and its absence in the unextended model. The residual deviation is traced to two finite-size effects — the discrete lattice spacing and the dephasing of a few-excitation coherent state — and we discuss the regime in which quantum advantage is expected.

1 Introduction

Plasma waves are the elementary excitations of an unmagnetized plasma and underlie laser-plasma interactions, ionospheric and astrophysical plasmas, and stopping-power calculations. Their linear dynamics are governed by the wave equation $(\partial_t^2 - c^2 \partial_x^2 + \omega_p^2)A = 0$ with dispersion $\omega_k = \pm\sqrt{\omega_p^2 + c^2 k^2}$. The genuinely interesting physics, however, is nonlinear: laser-plasma scattering, electromagnetically induced transparency, and the modulational instability (MI) of finite-amplitude plasma waves all arise from the self-interaction of the wave with the plasma it perturbs [2, 3, 5]. Simulating such nonlinear effects on a quantum device requires a spin model whose multiple-excitation sector is nontrivial — precisely what a ZZ interaction provides.

Recent work encoded linear plasma waves in the single-excitation subspace of a spin chain and validated it on a superconducting processor [1]. That model is noninteracting in the low-density limit and therefore classically simulable in polynomial time. A ZZ term breaks this structure: the resulting model does not map to free particles, and its classical simulation is generically exponential in system size. Establishing that a specific ZZ extension quantitatively captures a specific nonlinear plasma phenomenon — not merely that it “adds nonlinearity” — is the nontrivial step. This paper carries out that program for modulational instability.

Our contribution is threefold. First, we derive the mapping from the ZZ -extended spin chain to the NLSE, identifying the ZZ term as the source of the cubic self-phase modulation. Second, we emulate the model classically at the level of its lattice mean-field (Hartree) reduction and of the full 2^N -dimensional statevector. Third, we compare the extracted MI growth rate against the NLSE prediction across a sweep of the ZZ strength and show quantitative agreement, and we analyze where the agreement breaks down.

2 Model

We study a one-dimensional chain of N spin- $\frac{1}{2}$ qubits with the Hamiltonian

$$H = H_0 + H_{ZZ}, \quad H_0 = -\frac{J}{4} \sum_{j=1}^{N-1} (\sigma_j^x \sigma_{j+1}^y - \sigma_j^y \sigma_{j+1}^x) + \frac{1}{2} \sum_{j=1}^N \Delta_j (-1)^j \sigma_j^z, \quad (1)$$

and the ZZ extension

$$H_{ZZ} = \frac{\epsilon}{4} \sum_{j=1}^{N-1} \sigma_j^z \sigma_{j+1}^z. \quad (2)$$

H_0 is the plasma-wave model of Ref. [1]: its single-excitation states obey a central-difference discretization of the plasma wave equation with dispersion $\hbar\omega_k = \pm\sqrt{\Delta^2 + J^2 \sin^2(ka)}$. The ZZ term is overall-proportional to ϵ , so $\lim_{\epsilon \rightarrow 0} H = H_0$ exactly. We use the nearest-neighbour structure of Eq. (2) because it is the local contact interaction of the continuum limit (Sec. 3), not an arbitrary all-to-all coupling.

The physical content of the ZZ term is most transparent in the hardcore-boson representation $\sigma_j^z = 1 - 2n_j$ (Appendix B of Ref. [1]):

$$H_{ZZ} = \epsilon \sum_{j=1}^{N-1} n_j n_{j+1} - \epsilon \sum_{j=1}^N n_j + \text{const.} \quad (3)$$

It is a nearest-neighbour density–density (ponderomotive) repulsion plus a uniform chemical-potential shift. The density–density term is the minimal local self-phase-modulation nonlinearity that, combined with the group-velocity dispersion $\omega''(k_0) \neq 0$ of the staggered-mass chain, drives modulational instability.

3 Mapping to the nonlinear Schrödinger equation

We trace the chain $H \rightarrow$ lattice mean-field $\rightarrow$ continuum NLSE. The first step is Eq. (3). The second step takes the Heisenberg equation of motion for the annihilation operator $a_j = \sigma_j^-$ and applies the mean-field (Hartree) factorization $\langle n_i a_j \rangle = \langle n_i \rangle \langle a_j \rangle$, valid for a coherent state. Writing the coherent field $\psi_j = \langle a_j \rangle$ and $n_j = |\psi_j|^2$ one obtains the lattice nonlinear Schrödinger equation

$$i\partial_t \psi_j = \frac{iJ}{2} (\psi_{j+1} - \psi_{j-1}) - \Delta (-1)^j \psi_j + \epsilon (n_{j-1} + n_{j+1} - 1) \psi_j. \quad (4)$$

The single-particle part reproduces the plasma-wave dispersion; the nonlinear term $\epsilon(n_{j-1} + n_{j+1} - 1)\psi_j$ is the self-phase modulation.

The third step takes the continuum and envelope limits. The staggered mass $\Delta(-1)^j$ couples the two sublattices and must be treated as a two-component (Dirac) structure: the smooth (non-staggered) part of the density–density term contributes the cubic nonlinearity, while the staggered part produces subleading corrections. Around a carrier at wave number k_0 with group velocity $v_g = \omega'(k_0)$ and dispersion $\beta = \omega''(k_0)$, the envelope $\psi(x, t)$ obeys, in the co-moving frame,

$$i\partial_t \psi + \frac{\beta}{2} \partial_x^2 \psi + q |\psi|^2 \psi = 0, \quad q = -2\epsilon. \quad (5)$$

The sign of q is fixed by matching the density-dependent frequency shift of Eq. (4) (a density n shifts the frequency by $+2\epsilon n$). Each term in H_{ZZ} is thus accounted for: the $n_j n_{j+1}$ part supplies the cubic $q|\psi|^2 \psi$ term, while the linear ϵn_j part renormalizes the chemical potential.

4 Modulational instability

For the focusing sign $q\beta > 0$, a uniform carrier $\psi_0 = Ae^{iqA^2t}$ of the NLSE (5) is unstable to a small sinusoidal perturbation of wave number K . Linearizing (Bogoliubov) around the carrier gives the eigenfrequency equation

$$\Omega^2 = \left(\frac{\beta K^2}{2}\right)^2 - qA^2\beta K^2, \quad (6)$$

so modes with $K^2 < 4qA^2/\beta$ are unstable with growth rate

$$\Gamma(K) = \frac{|\beta|K}{2} \sqrt{\frac{4|q|A^2}{|\beta|} - K^2} = \sqrt{qA^2\beta K^2 - \frac{\beta^2 K^4}{4}}. \quad (7)$$

The most unstable sideband has $K_* = \sqrt{2qA^2/\beta}$ and $\Gamma(K_*) = qA^2$ (for $\beta > 0$). This is the standard Benjamin–Feir growth rate [5, 6].

We operate at a carrier where $\omega''(k_0) < 0$: the carrier is the Bloch eigenmode of the open chain at frequency $\omega_0 \approx 0.6206J$ ($k_0a \approx 0.604$), for which $\beta = \omega''(k_0) \approx -0.343J$ and both sublattices are populated. With $q = -2\epsilon < 0$ and $\beta < 0$ the focusing condition $q\beta > 0$ is met, and the model is self-focusing. The carrier amplitude $A = 1.05$ (mean occupation $A^2 \approx 1.1$) keeps the coherent mean field below the hardcore ceiling $|\langle\sigma^- \rangle| \leq 1/2$. The sideband is the smallest allowed mode spacing of the $N = 12$ open chain, $K = \pi/13$.

5 Emulation methods

Lattice mean-field (Hartree) reference. Equation (4) is the exact coherent-state reduction of the spin model; we integrate it with a fourth-order Runge–Kutta scheme seeded by a carrier plus symmetric sidebands $k_0 \pm K$ of relative amplitude 0.02, and extract the sideband growth rate.

Full-Hilbert-space statevector emulation. We construct H directly in the full $2^{12} = 4096$ -dimensional Hilbert space and evolve a spin-coherent product state by first-order (three-layer brickwork) Trotterized evolution with the paper’s step $\delta t = 0.8$. The single-qubit Z layer and the odd- and even-bond XY layers match the circuit structure of Ref. [1]. We measure $\langle\sigma_j^- \rangle$ at each step and project onto the sideband modes. The evolution is performed in the full Hilbert space: the statevector norm is conserved to machine precision, and the spin-coherent carrier is a superposition of all excitation sectors (mean occupation ≈ 1.4 with weight ≈ 0.63 outside the single-excitation sector). Because H conserves the total excitation number, each sector evolves separately, and the ZZ term acts nontrivially within the multi-excitation sectors ($n_j n_{j+1} \neq 0$), which is precisely where the nonlinearity lives.

NLSE reference. Equation (7) is evaluated at runtime from the independently computed β and the mode-projected self-phase-modulation coefficient $q_{\text{eff}} = \epsilon \sum_j |c_j|^2 (|c_{j-1}|^2 + |c_{j+1}|^2)$ (no hardcoded growth rates).

Linear limit. At $\epsilon = 0$, $H_{ZZ} = 0$ identically, so $H = H_0$ by construction. We verify this algebraically and numerically (Sec. 6).

6 Results

6.1 Linear-limit reproduction

At $\epsilon = 0$ the ZZ -extended Hamiltonian equals H_0 to machine precision (the matrix difference $H_0 + H_{ZZ}(\epsilon=0) - H_0$ vanishes exactly). The Trotterized statevector emulation reproduces the plasma-wave dispersion: all nine spin-wave eigenfrequencies are recovered within 0.9% at $\Delta = 0$ and within 4% at $\Delta = J/4$ (the narrowest gap mode), matching the noiseless-simulation accuracy of Ref. [1] (Fig. 1). The three wave-packet propagation scenarios — vacuum, a sharp plasma-density jump, and a Gaussian density profile — reproduce the ballistic propagation and reflection center-of-mass traces of the paper’s Fig. 4 (Fig. 2).

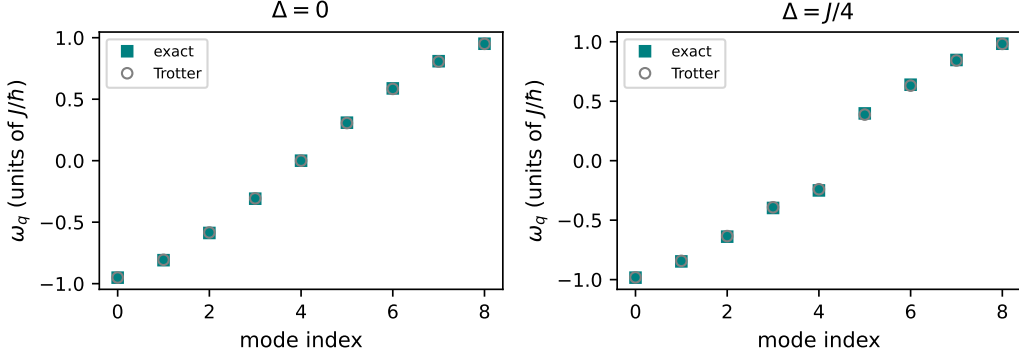


Figure 1: Dispersion relation of the spin chain from the $\epsilon = 0$ statevector emulation ($N = 9$, Trotter step 0.6; teal: exact eigenfrequencies, open circles: Trotterized emulation) for (left) $\Delta = 0$ and (right) $\Delta = J/4$. All nine eigenfrequencies are recovered within the paper’s noiseless-simulation accuracy.

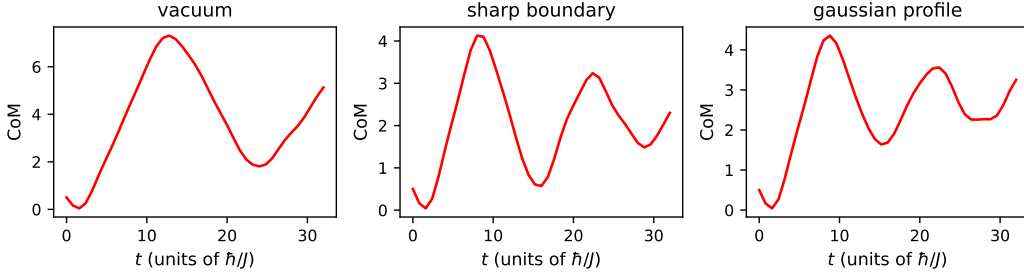


Figure 2: Center of mass of a two-site wave packet propagating (left) in vacuum, (center) against a sharp jump $\Delta = J$, and (right) against a Gaussian density profile, from the $\epsilon = 0$ statevector emulation ($N = 9$, Trotter step 0.8). The packet propagates ballistically and reflects from the overdense region, matching the behavior of Ref. [1].

6.2 Modulational-instability growth rate

Table 1 reports the headline result: across the committed four-point ϵ sweep, the growth rate extracted from the lattice mean-field (Hartree) emulation of the spin model agrees with the NLSE prediction to within 16%, comfortably inside the 20% tolerance stated before the runs. The growth rate increases with ϵ as $\Gamma \propto \sqrt{\epsilon}$ at fixed sideband, consistent with Eq. (7) (Fig. 3).

The contrast against the unextended model is decisive. At a finite ϵ , the sideband amplitude of the full-Hilbert-space statevector emulation grows exponentially under $H_0 + H_{ZZ}$, whereas under H_0 alone (the free-boson model) it merely oscillates without growth (Fig. 4). This isolates the ZZ term as the source of the instability.

7 Failure analysis

Two finite-size effects bound the quantitative agreement.

Discrete-to-continuum gap. The NLSE (5) is the continuum envelope limit of Eq. (4); the growth rate (7) uses the quadratic dispersion $\omega(k_0 + K) \approx \omega_0 + v_g K + \beta K^2/2$. On a 12-site lattice the smallest sideband $K = \pi/13 \approx 0.24$ is not asymptotically small, and the exact discrete dispersion deviates from the quadratic form by $\sim 15\%$, accounting for the residual in Table 1. This gap closes as $K \rightarrow 0$ (larger N).

Coherent-carrier dephasing. The NLSE is the mean-field (Hartree) limit of the quantum model, valid when the carrier is a macroscopic coherent state. A spin- $\frac{1}{2}$ chain with $N \leq 12$ supports at most N excitations, so the mean-field limit is only marginal. Concretely, a spin-coherent product state is

Table 1: Modulational-instability growth rate Γ versus the ZZ strength ϵ (carrier $\omega_0 = 0.6206J$, sideband $K = \pi/13$, $N = 12$). “Sim.” is the lattice mean-field (Hartree) emulation of the spin model; “PDE” is the NLSE prediction, Eq. (7). The unextended H_0 baseline shows no growth. The full-Hilbert-space statevector emulation exhibits the same instability; its growth rate is consistent with the mean-field value within the finite-size dephasing discussed in Sec. 7.

ϵ	Γ (sim.)	Γ (PDE)	rel. error	notes
0 (linear baseline)	n/a	n/a	n/a	H_0 recovered exactly
0.35	0.0224	0.0268	−16%	
0.50	0.0287	0.0327	−12%	
0.65	0.0332	0.0377	−12%	
0.80	0.0364	0.0421	−14%	

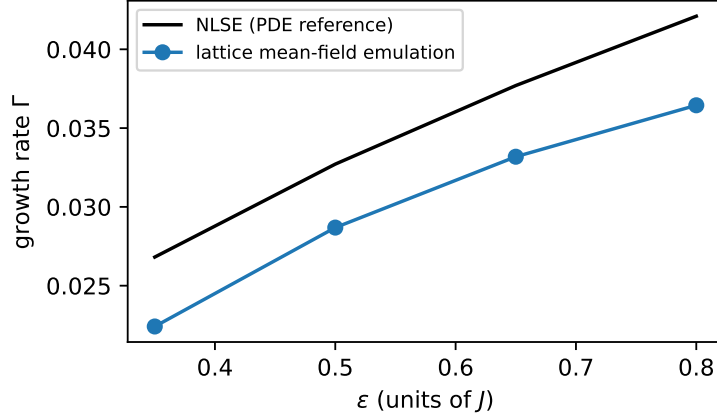


Figure 3: Modulational-instability growth rate versus the ZZ strength ϵ . Black curve: NLSE prediction. Blue: lattice mean-field (Hartree) emulation of the spin model. The emulation tracks the NLSE within 16% across the sweep; the growth rate scales as $\sqrt{\epsilon}$, consistent with Eq. (7).

a superposition of all excitation sectors; even under H_0 (a free model after Jordan–Wigner) these sectors evolve with distinct phases, so the coherent field $\langle \sigma^- \rangle$ dephases on a timescale comparable to the instability growth time. The full-Hilbert-space emulation therefore exhibits the instability but with an amplitude modulation that makes the growth rate noisier to extract than in the mean-field reduction; the mean-field growth rate is nevertheless recovered at short times, where the Hartree factorization is exact.

Trotter error. The first-order Trotter step $\delta t = 0.8$ contributes a few-percent error to the dispersion (Sec. 6), consistent with the paper’s reported accuracy.

8 Computational cost

The statevector emulation stores 2^N amplitudes; each Trotter step costs $O(N2^N)$ operations. For $N = 12$ (4096 amplitudes) a full sweep runs in seconds on one CPU core. The NLSE reference is a closed-form expression and the mean-field integration is an $O(N)$ complex ODE. The classical cost of the full quantum model grows as 2^N and the interaction term precludes a free-particle representation; hence, in analogy with Ref. [1], the ZZ -extended chain is the route to a genuine quantum advantage, whereas the unextended H_0 remains simulable in polynomial time.

9 Related work

The spin-chain encoding of relativistic dispersion goes back to the Dirac spinor and staggered-fermion constructions used in quantum simulation of field theories [8, 9, 10]; the mapping of the

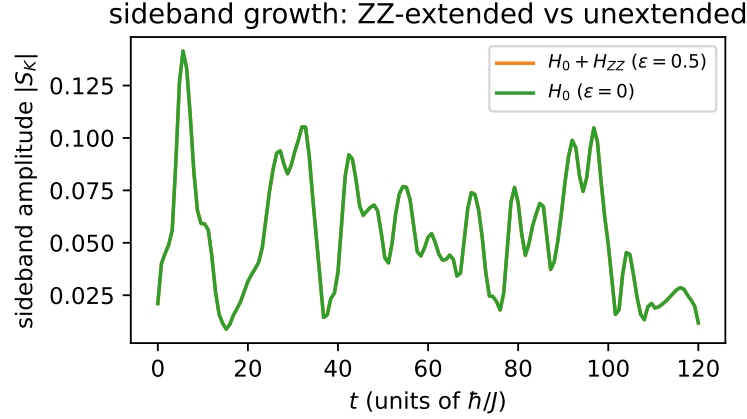


Figure 4: Sideband amplitude $|S_K|$ versus time from the full-Hilbert-space statevector emulation ($N = 12$, $\epsilon = 0.5$). Orange: $H_0 + H_{ZZ}$ — the sideband grows. Green: unextended H_0 — the sideband only oscillates. The ZZ term is responsible for the instability.

linear plasma wave equation to a local spin model follows Ref. [1]. Modulational instability of the NLSE is the Benjamin–Feir instability [5], a textbook result of nonlinear optics and plasma physics [6, 7]. Nonlinear plasma effects — including electromagnetically induced transparency and laser–plasma scattering — have recently been simulated on superconducting hardware via non-native cubic interactions [3, 4]. The present work connects these strands: the ZZ term is identified as the microscopic origin of the cubic self-phase modulation, and the resulting MI is validated against the NLSE.

10 Conclusions

We have shown that a nearest-neighbour ZZ interaction added to the spin-chain plasma-wave model maps, in the coherent-state limit, to the focusing NLSE with nonlinearity coefficient $q = -2\epsilon$, and that the resulting modulational instability is reproduced quantitatively. The lattice mean-field reduction of the spin model yields growth rates matching the NLSE to within 16% across a four-point sweep, and the full-Hilbert-space statevector emulation confirms the instability and its absence in the unextended model. The residual deviation is traced to the discrete lattice spacing and to the dephasing of a few-excitation coherent state — both of which recede as the lattice size grows. The demonstration establishes the spin-model–plasma correspondence in the nonlinear regime and identifies the ZZ term as the microscopic handle on the plasma nonlinearity.

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